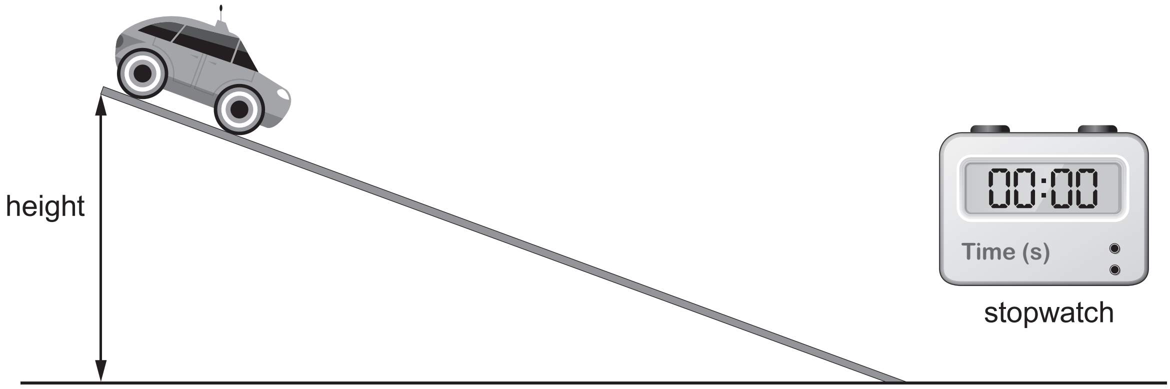
Examiner only

4. Two students carry out an experiment with a toy car and a 2.50 m long piece of track.



They investigate how changing the height at one end of the track affects the time taken for the toy car to travel down 2.50 m of the track. One student releases the car and the other uses a stopwatch to measure the time. They do this 3 times for each height.
Their results are shown in the table.

Height (cm)	Distance travelled (m)	Time (s)			
		Result 1	Result 2	Result 3	Mean
10	2.50	4.0	4.1	4.0	4.0
20	2.50	2.9	3.3	3.1	3.1
30	2.50	2.5	2.7	2.4	2.5
40	2.50	2.1	2.0	2.3	2.1
50	2.50	2.0	1.8	1.9	1.9

(a) Identify a controlled variable in the table. [1]

....

Examiner only

(b) (i) Use the equation:

$$\text{mean speed} = \frac{\text{distance travelled}}{\text{mean time}}$$

to calculate the mean speed of the toy car when the slope is set at a height of 10 cm. [2]

Mean speed = ... m/s

(ii) I. Describe how the mean time changes as the height increases by 10 cm steps. [2]

....

....

II. Describe how the mean speed changes as the height increases. [1]

....

(c) (i) One student says, “The most repeatable data is for a height of 50 cm”. Explain why this statement is **incorrect** and write a similar correct statement. [2]

....

....

....

(ii) Explain why using a timer connected to light gates positioned at the start and end of the 2.50 m track will improve the results. [2]

....

....

....

10